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Rice Yield Estimation Based on Machine Learning Applied to UAV Remote Sensing Data

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Highlights

What are the main findings?

- Multi-year temporal validation of UAV-based machine learning models across three growing seasons (2023–2025) demonstrated robust yield prediction.
- Comprehensive ML benchmarking revealed that tree-based models (XGBoost and Random Forest: R^2 = 0.85) substantially outperformed Linear Regression and neural networks.
- The EVI and NDRE at the reproductive stage were the most consistent predictors of rice grain yield across 300 iterations.

What are the implications of the main findings?

- UAV-based ML can support accurate in-season rice yield estimation and improve nitrogen management.
- The strong independent validation in 2025 suggests the models have practical potential for operational use across growing seasons.

Abstract

Accurate in-season rice (*Oryza sativa* L.) yield prediction is crucial for improved nitrogen management and climate-smart decision making, yet rigorous comparative benchmarking of machine learning (ML) models using multi-temporal UAV spectral data with independent temporal validation remains limited. This study systematically evaluated four ML algorithms (Random Forest, XGBoost, Neural Network, and Linear Regression) and two Bayesian model averaging ensembles for rice yield prediction using UAV multispectral imagery. Field experiments spanning three growing seasons (2023–2025) at Louisiana State University comprised 9–10 varieties and six nitrogen rates (0–235 kg N ha^{-1}; 576 plots). Vegetation indices and spectral bands from three growth stages were extracted as predictors. Models were compared using 300 random train–test iterations with systematic hyperparameter optimization, followed by independent validation on 2025 data. Among the individual models, XGBoost achieved the highest internal accuracy (R^2 = 0.87, RMSE = 0.85 t ha^{-1}), substantially outperforming Linear Regression (R^2 = 0.66, RMSE = 1.32 t ha^{-1}), while reduced BMA reached R^2 = 0.89. XGBoost demonstrated robust temporal generalization (R^2 = 0.62, NRMSE = 8.47%) despite environmental variation. The Enhanced Vegetation

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Index and Normalized Difference Red Edge at 90 days after planting (reproductive stage) were the most stable predictors across 300 iterations. Tree-based ML models substantially outperform traditional linear approaches, providing reliable pre-harvest yield forecasting for operational precision rice production.

Keywords: rice yield prediction; UAV remote sensing; machine learning; multispectral imagery; NDVI; NDRE; precision agriculture; nitrogen management

1. Introduction

Rice (*Oryza sativa* L.) is one of the world's most important staple crops, providing food for over 3.5 billion people and serving as a primary calorie source for approximately half of the global population [1,2]. Global rice production exceeded 520 million metric tons in 2023, cultivated across more than 165 million hectares, mostly in Asia, where 90% of production and consumption occurs [2]. The productivity and yield of rice are linked to food security, particularly in Asia and other rice-consuming regions [2]. However, meeting the projected demand for rice, which is expected to increase by 25% by 2050, requires substantial improvements in crop management practices [3,4].

An accurate estimation of rice yields before harvest is crucial for guiding policy decisions, stabilizing market prices, and ensuring adequate food supply at national and global scales [5]. In precision agriculture, early and accurate yield predictions enable optimized resource allocation, particularly for nitrogen fertilization, which is the biggest factor affecting rice productivity [6].

Traditionally, crop yields have been estimated through destructive ground-based measurements and farmer surveys, which are labor-intensive, time-consuming, and often infeasible at large scales [7]. Such methods involve clipping and threshing plants from sample plots to extrapolate yields, a process that is time-consuming and prone to sampling errors. Moreover, destructive sampling cannot be repeated throughout the season without sacrificing yield, precluding this continuous loop of reduced yield.

Process-based crop growth simulation models (e.g., DSSAT [8], APSIM [9], and WOFOST [10]) offer an alternative by predicting yield from meteorological, soil, and management inputs. These models simulate growth processes and have been proven valuable [8,11]. However, their operational deployment faces significant challenges: they require extensive localized data and site-specific calibration, including detailed weather, soil properties, and crop management information [12]. This dependence on high-quality input data and the models' complexity can limit their usability and accuracy, especially when deployed beyond well-characterized research settings [12]. Moreover, process-based models can be computationally intensive and may not readily accommodate unforeseen stress factors without additional calibration [13]. Statistical estimation models provide an alternative empirical approach to yield predictions without simulating physiological processes. These models establish relationships between crop productivity and observable environmental or agronomic variables. The integration of remote sensing data into statistical frameworks has substantially enhanced their predictive power and spatial applicability [14].

Remote sensing technologies have emerged as powerful tools for large-scale, non-destructive crop monitoring, opening new avenues for yield estimation [14]. Since the 1970s, satellite remote sensing has been used to predict crop production over wide areas [15,16]. Satellite imagery provides consistent coverage and has enabled regional yield forecasts using vegetation indices (VIs), thermal signals, and other indicators. For example, MODIS and Sentinel-2 imagery have been successfully coupled with climate data to predict wheat

and rice yields at provincial and national scales [17,18]. However, satellite data are often constrained by coarse spatial resolution, cloud cover, and long revisit intervals, which hinder precise and timely plot-level monitoring [19].

Unmanned aerial vehicles (UAVs) have revolutionized high-resolution crop monitoring, overcoming many satellite limitations through on-demand deployment and superior spatial–temporal resolution [13,20]. UAV platforms can acquire centimeter-level imagery at user-defined intervals, enabling the detection of within-field variability and precise phenological timing [21,22]. More importantly, UAVs operate below cloud cover, ensuring data acquisition during critical windows that may be obscured in satellite imagery. UAV systems accommodate diverse sensors, including red–green–blue (RGB), multispectral, hyperspectral, thermal, and light detection and ranging (LiDAR), providing flexibility for capturing a wide range of crop properties [12]. This versatility is particularly valuable in rice systems, where smallholder fields, heterogeneous management practices, and frequent cloud cover during monsoons necessitate flexible, high-resolution precision monitoring.

Among UAV-derived data products, VIs calculated from multispectral imagery have become the standard approach for crop vigor assessment and yield forecasting [23]. Indices such as the normalized difference vegetation index (NDVI) correlate strongly with the leaf area index (LAI), chlorophyll content, biomass accumulation, and ultimately grain yield across growth stages [20,24]. However, using individual indices exhibits certain limitations: the NDVI saturates in dense canopies, reducing sensitivity significantly, especially when yield formation is most crucial [25,26]. For rice, the NDVI is most informative during early growth but tends to plateau around heading, which can reduce its usefulness for predicting grain filling and final yield.

To address this limitation, alternative indices have been developed that leverage different spectral regions and mathematical formulations. The Enhanced Vegetation Index (EVI) incorporates blue-band reflectance for atmospheric correction and a soil-adjustment factor, maintaining sensitivity when red-NIR reflectance saturates. Normalized Difference Red Edge (NDRE) [27] utilizes red-edge wavelengths (705–750 nm), which exhibit a steeper response gradient to chlorophyll concentration than red-band reflectance, enabling the detection of canopy nitrogen variations even at high biomass. The Green NDVI (GNDVI) [28] substitutes green for red reflectance, penetrating deeper into multi-layered canopies and maintaining sensitivity during canopy closure. These indices capture complementary canopy properties, and recent research demonstrates that multi-index, multi-temporal approaches substantially improve yield estimation compared to single-index methods [29–31].

In parallel with advances in UAV remote sensing, machine learning (ML), a subfield of artificial intelligence (AI) focused on learning patterns from data without explicit programming, has emerged as the dominant approach for translating spectral data into yield predictions. Unlike simple empirical regression models that mostly assume linear relationships, ML algorithms can capture complex, nonlinear relationships between predictors and crop yields [12,31]. Early applications of ML in yield forecasting date back to the use of neural networks (NNs) [32,33], support vector machines (SVMs) [30,31], and decision trees like Random Forest regression (RF) [34–36] on satellite-derived indices in the 1990s and 2000s [13]. In recent years, the explosion of "big data" and improved computing resources has solidified ML as a popular approach for yield modeling, often outperforming or supplementing traditional regression models [37].

The contemporary ML techniques for yield prediction cover diverse algorithmic approaches, each with distinct strengths. RF has become one of the most widely applied methods for UAV-based yield estimation, valued for handling high-dimensional feature spaces, capturing non-linear relationships, and providing built-in importance metrics through ensemble learning [12]. Gradient boosting methods, particularly Extreme Gradient

Boosting (XGBoost), have also shown similar performance by sequentially correcting prediction errors and incorporating advanced regularization to prevent overfitting. Deep learning approaches, especially NNs, can extract spatial features from raw imagery and are equally being used along with ML models, but they have higher data requirements and intense computation [6,13].

Critically, it is notable that model complexity does not always guarantee superior performance. Studies comparing algorithms on identical datasets have found that simpler models (Linear Regression, RF) can match or even exceed deep learning approaches when feature engineering is appropriate and sample sizes are limited [38]. This finding highlights the importance of feature richness and appropriate model choice tailored to data characteristics, rather than defaulting to the most complex approach [13].

Ensemble methods that combine predictions from multiple models have also gained attention for potentially improving accuracy and robustness beyond individual models being used. Bayesian model averaging (BMA), for instance, weighs predictions from diverse models according to their estimated performance, theoretically capitalizing on complementary strengths while mitigating individual weaknesses [39]. However, it is still underexplored in agricultural remote sensing.

Despite these advances, existing UAV-based rice yield prediction studies exhibit several limitations. First, most studies lack systematic algorithm comparison: they typically employ a single ML model or a limited comparison [40,41]. For example, Zhou et al. (2017) [40] focused on VIs' selection without systematic algorithm comparison, while Mia et al. (2023) [41] applied multimodal deep learning as a single approach. Without standardized hyperparameter optimization and consistent evaluation protocols, reported performance metrics are often incomparable across studies. Second, independent temporal validation remains scarce: the majority of studies rely on cross-validation within a single growing season [42,43], which cannot demonstrate whether models have learned genuine biological relationships or merely memorized year-specific patterns. Wang et al. (2021) [42] and Duan et al. (2019) [43] both reported strong predictions, leaving temporal generalizability unassessed. Third, stability assessment is largely absent: performance metrics are typically reported from single train–test splits [41,43], providing limited information about prediction variance and model robustness. Fourth, feature importance analysis, when performed, is often derived from single model fits, without evaluating stability across multiple data partitions, which may lead to importance scores that are sensitive to specific training splits [44]. These limitations collectively prevent the development of robust, transferable forecasting systems for precision rice management.

This study addresses these limitations through a comprehensive, methodologically rigorous framework for UAV-based rice yield prediction. Our approach makes five key innovations compared to existing research. First, we implement systematic algorithm benchmarking: four diverse ML algorithms plus two BMA ensemble configurations are trained and evaluated on identical datasets with consistent hyperparameter optimization, enabling robust performance comparison and the identification of optimal modeling strategies. Second, we demonstrate true temporal generalization: models developed on 2023–2024 data undergo independent validation on an entirely separate 2025 growing season with substantially different environmental conditions, providing robust evidence of model transferability beyond cross-validation. Third, we employ data-driven feature discovery: rather than pre-selecting indices, we systematically evaluate multiple spectral features at three critical growth stages, enabling a comprehensive assessment of the full predictive feature space. Fourth, we quantify prediction stability: iterative evaluation across 300 random data partitions provides confidence intervals for all performance metrics, characterizing model robustness and reliability for operational deployment. Fifth, we

conduct robust feature importance analysis across all 300 train–test iterations, quantifying not only which predictors are most influential but also the consistency and stability of their importance rankings across different data partitions. This approach distinguishes genuinely important features that consistently contribute to predictions from those that appear influential by chance in single model fits, providing reliable guidance for optimal sensor deployment and observation timing in operational settings.

The study leverages three years of UAV multispectral data collected under diverse varietal composition (14 cultivars) and nitrogen management regimes (0–235 kg N ha^{-1}), representing realistic agronomic heterogeneity. The findings provide actionable insights for selecting appropriate modeling strategies, optimal observation timing, and critical predictor variables for operational UAV-based yield forecasting, thereby supporting precision rice management and food security planning.

2. Materials and Methods

2.1. Study Area and Experimental Design

Field experiments were conducted at the Louisiana State University AgCenter Rice Research Station in Rayne, Louisiana, USA (30°14′50.6″N, 92°21′2.7″W) during the 2023, 2024, and 2025 growing seasons (Figure 1). The region experiences a humid subtropical climate with mean annual temperatures of 19.8 °C and precipitation of 1540 mm, providing optimal conditions for rice cultivation. The experimental site is characterized by Crowley silt loam soils (fine, smectitic, thermic Typic Albaqualfs).

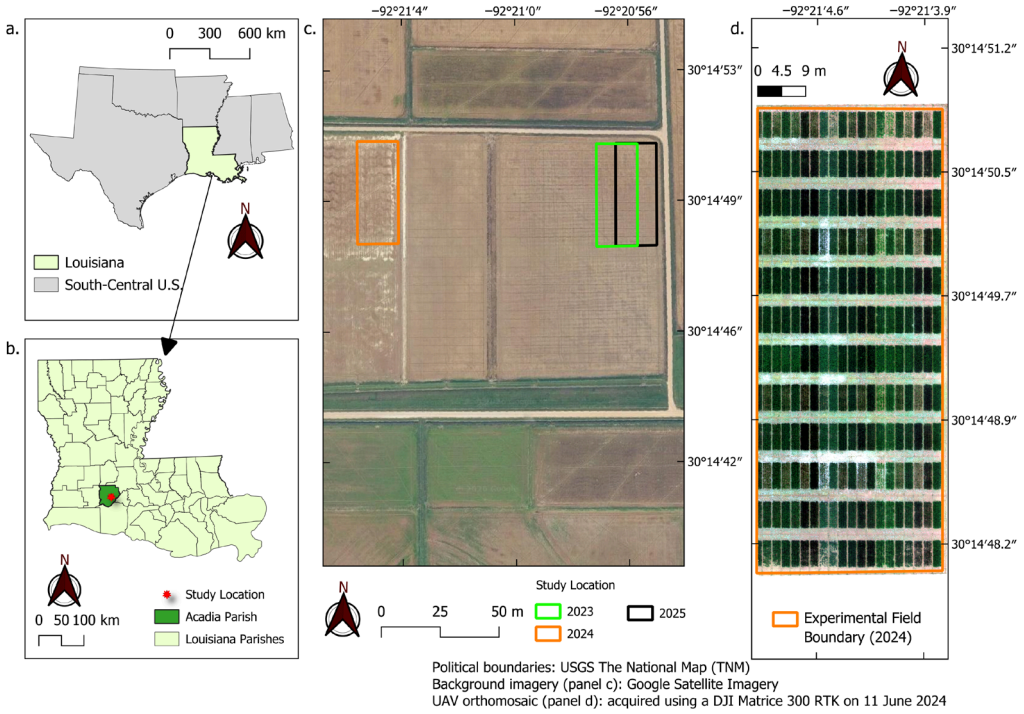

Figure 1. The location of the experimental field at LSU AgCenter Rice Research Station, Rayne, LA, showing (**a**) regional context, (**b**) Acadia Parish, (**c**) aerial view with year-specific areas, and (**d**) plot arrangement.

Pre-plant–soil sampling revealed inter-annual variation in certain soil properties (Table 1), particularly pH, likely reflecting natural spatial heterogeneity characteristic of this soil series. Similar variation was observed in extractable phosphorus and exchangeable calcium. Despite these differences, all measured values remained within agronomically acceptable ranges for rice production. To minimize nitrogen carryover effects, the experimental field was left fallow between growing seasons, and the randomized complete block

design (RCBD) with four replications ensured that spatial variation was distributed evenly across treatments within each season.

Table 1. Pre-plant top-soil nutrient properties (0–15 cm depth).

Year	OM (g kg^{-1})	pH	Ca (mg kg^{-1})	Mg (mg kg^{-1})	K (mg kg^{-1})	Na (mg kg^{-1})	P (mg kg^{-1})	S (mg kg^{-1})	Zn (mg kg^{-1})	Cu (mg kg^{-1})
2023	14.7	7.4	1563	301	58	103	20	8.1	7.3	1.49
2024	13.5	6.5	1203	243	52	113	16	5.6	8	1.3
2025	14.7	6.6	1553	307	70	140	43	8.9	10	1.7

In 2023, 10 rice varieties were evaluated, resulting in a total of 240 plots, while in 2024, 9 varieties were evaluated, resulting in 216 plots (Table 2). Across 2023 and 2024, 456 plots were established (Table 2, Figure 2), with each plot serving as one observation for modeling. Data from these two seasons were used for model development with an 80/20 train–test split for internal validation. In 2025, five varieties that had been evaluated in at least one previous season were selected for external validation, resulting in 120 independent validation plots. Each plot measured 4.67 m (width) × 4.88 m (length) and accommodated seven drill-seeded rows at 20.3 cm row spacing.

Table 2. Rice genotypes used for each year's trial.

2023 Varieties	2024 Varieties	2025 Varieties
CLHA03	CLHA03	CLHA03
PVL03	PVL04	PVL04
CLL16	Taurus	Taurus
CLL19	Fitzgerald	Fitzgerald
Avant	23DGL-003PV	23DGL-003PV
DGM004	DGM004	
DGL2065	DGL2065	
CLL18	CLL18	
Addi Jo	Addi Jo	
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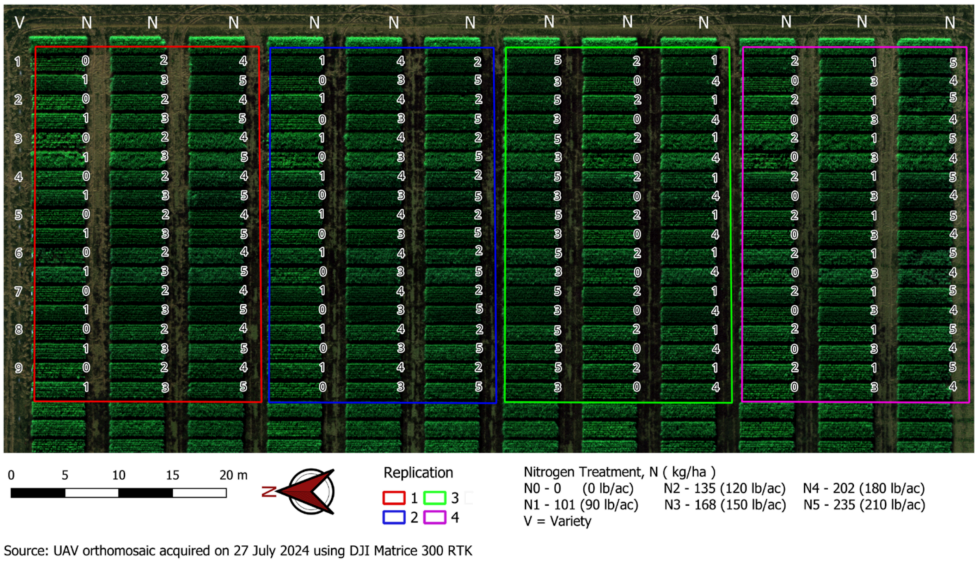

Figure 2. The experimental layout of the variety × nitrogen field trial, overlaid on a UAV Orthomosaic image (27 July 2024). The plots denote rice varieties (V: 1–9) and nitrogen treatments (N: 0–5; 0–235 kg N ha^{-1}). The colored borders indicate the four replication blocks (Replications 1–4).

Nitrogen fertilization treatments consisted of six rates: 0, 101, 135, 168, 202, and 235 kg N ha^{-1}, applied as urea (46% N) across all years (Table 3). Here, 0 kg N ha^{-1} treatment served as the control, while 168 kg N ha^{-1} represents the regional recommended rate.

Table 3. A synchronized timeline of field operations and UAV data collection for 2023–2025 growing seasons. Minor variations in calendar dates between years reflect planting date differences.

Event	2023 (DAP)	2024 (DAP)	2025 (DAP)
Planting	8 March (0)	13 March (0)	10 March (0)
Emergence	28 March (20)	27 March (14)	24 March (14)
UAV Flight 1	18 April (41)	1 May (49)	30 April (51)
N Fertilization	3 May (57)	15 May (63)	29 April (50)
Flooding	4 May (58)	16 May (64)	30 April (51)
Panicle initiation	29 May (83)	30 May (78)	19 May (70)
UAV Flight 2	8 June (92)	11 June (90)	5 June (87)
50% Heading	28 June (112)	27 June (106)	21 June (103)
UAV Flight 3	10 July (124)	17 July (126)	17 July (129)
Harvest	27 July (140)	31 July (140)	22 July (134)

Standard agronomic management practices for drill-seeded irrigated rice were followed in both years, including timely irrigation (flush and permanent flood), weed and pest management as recommended for the region [45]. Planting, emergence, start of irrigation, and harvest date information for the study are provided in Table 3.

2.2. UAV Data Acquisition and Processing

UAV remote sensing was employed to collect multispectral and RGB imagery data of the trials at key growth stages. This study used a DJI Matrice 300 RTK UAV (DJI, Shenzhen, China) equipped with a MicaSense Red Edge-MX multispectral camera (MicaSense, Seattle, WA, USA) and a DJI Zenmuse H20 RGB camera (DJI, Shenzhen, China). The MicaSense Red Edge-MX sensor captures five spectral bands with type and central wavelength as follows: blue (475 nm), green (560 nm), red (668 nm), red edge (717 nm), and near-infrared (840 nm). In each growing season, 10 mapping mission flights were conducted, out of which 3 flights correlated with critical growth stages: (1) early vegetative (40–49 days after planting {DAP}), (2) reproductive (90–92 DAP), and (3) grain filling (124–126 DAP) were included in the datasets for yield estimation (Table 3).

The flight missions were set at 45 m above ground level with data capture methodology following the requirement for UAV photogrammetry. At this altitude, the MicaSense RedEdge-MX multispectral camera achieved a ground sampling distance (GSD) of approximately 2.5 cm per pixel. Ground control points (GCPs) data were collected using EMLID Reach RS2 Global Navigation Satellite System Real Time Kinematic (GNSS RTK)((EMLID Ltd., Budapest, Hungary) instrument with a base correction involving NASA OPUS (v5.0.0; https://www.ngs.noaa.gov/OPUS/, accessed on 15 March 2023) to ensure planimetrically accurate and consistent final orthomosaic maps across all UAV data acquisitions [46].

Raw images from each flight were processed into calibrated orthomosaic images using Pix4D Mapper (v4.7.5, Pix4D S.A., Lausanne, Switzerland) [47]. Radiometric calibration (using MicaSense reflectance panel captures) was performed in Pix4D Mapper for each spectral band to mitigate varying solar radiation and aircraft tilt at the time of data acquisition. The resulting band-specific orthomosaics were then stacked into multiband composite maps using the Geospatial Data Abstraction Library (GDAL, v3.11) [48]. All spatial data were projected to Universal Transverse Mercator (UTM) zone 15 based on the World Geodetic System (WGS) 1984 ensemble datum (EPGS: 32615).

Raster calculator in QGIS (v3.28.1) [49] was used to calculate five VIs sensitive to crop vigor and N status from the multispectral data (Table 4). The NDVI was used for its strong correlation with the leaf area index (LAI) and biomass [24]. The GNDVI was used for its enhanced sensitivity to chlorophyll content compared to the NDVI, particularly in dense canopies [28]. The NDRE was chosen for its ability to detect subtle variations in canopy nitrogen concentration through red-edge reflectance, which is less prone to saturation than the NDVI in high-biomass conditions [27]. The EVI was incorporated to mitigate atmospheric and soil background effects while maintaining sensitivity under dense canopy conditions [50]. The normalized green–red difference index (NGRDI) was included as a complementary index for assessing canopy greenness and crop vigor using visible wavelengths [51]. These indices leverage different spectral regions to capture complementary aspects of canopy structure, chlorophyll content, and photosynthetic activity (Table 4). The mean values of the VIs for each plot were extracted by overlaying plot boundary polygons on the raster layers, providing plot-level UAV features that could be matched with ground truth yield data for modeling.

Table 4. Vegetation indices calculated from UAV multispectral imagery. Equations show band combinations where NIR = near-infrared (840 nm), RE = red edge (730 nm), R = red (650 nm), G = green (560 nm), and B = Blue (475 nm). Constants in the EVI equation are standard coefficients for atmospheric and soil background correction.

Vegetation Index		Equation	Ref.
Name	**Description**		
NDRE	Normalized Difference Red Edge	$\frac{(NIR-RE)}{(NIR+RE)}$	[27]
GNDVI	Green Normalized Difference Vegetation Index	$\frac{(NIR-G)}{(NIR+G)}$	[28]
NDVI	Normalized Difference Vegetation Index	$\frac{(NIR-R)}{(NIR+R)}$	[24]
EVI	Enhanced Vegetation Index	$\frac{2.5(NIR-R)}{(NIR+6R-7.5B+1)}$	[50]
NGRDI	Normalized Green–Red Difference Index	$\frac{(G-R)}{(G+R)}$	[51]

2.3. Data Integration and Preprocessing

2.3.1. Measurement Timeline and Synchronization

Experimental activities followed a coordinated timeline (Table 3) designed to capture the yield formation across the growth cycle. Planting occurred in early March, and nitrogen fertilizer was applied at 50–63 DAP across the years. From the ten UAV flights conducted each season, imagery acquired at 45, 90, and 125 DAP was used in this study, representing key phenological stages (tillering, heading, and grain filling). Final grain yield was measured at harvest maturity by harvesting each entire plot using a Wintersteiger Delta plot combine harvester (Wintersteiger AG, Ried im Innkreis, Austria) equipped with a HarvestMaster H2 Grain Gauge (Juniper Systems, Logan, UT, USA). Grain yield was reported at 12% moisture content.

2.3.2. Dataset Preparation

The overall workflow for building and evaluating the rice yield prediction models is illustrated in Figure 3. The final dataset integrated 47 input features and one target variable (grain yield, t ha^{-1}). The input features comprised: 15 UAV-derived VIs (5 indices × 3 time points: 45, 90, and 125 DAP); 15 spectral band reflectances (5 bands × 3 time points); nitrogen application rate (continuous: 0–235 kg N ha^{-1}); and categorical variables (year and variety) encoded using one-hot encoding. The year (2 categories: 2023 and 2024) was converted into 2 binary indicators, and the variety

(14 unique cultivars across 2023–2024) was converted into 14 binary indicators, with no reference category dropped (full dummy encoding). This encoding approach is appropriate for tree-based algorithms, which are robust to collinearity, but may introduce redundancy in linear models.

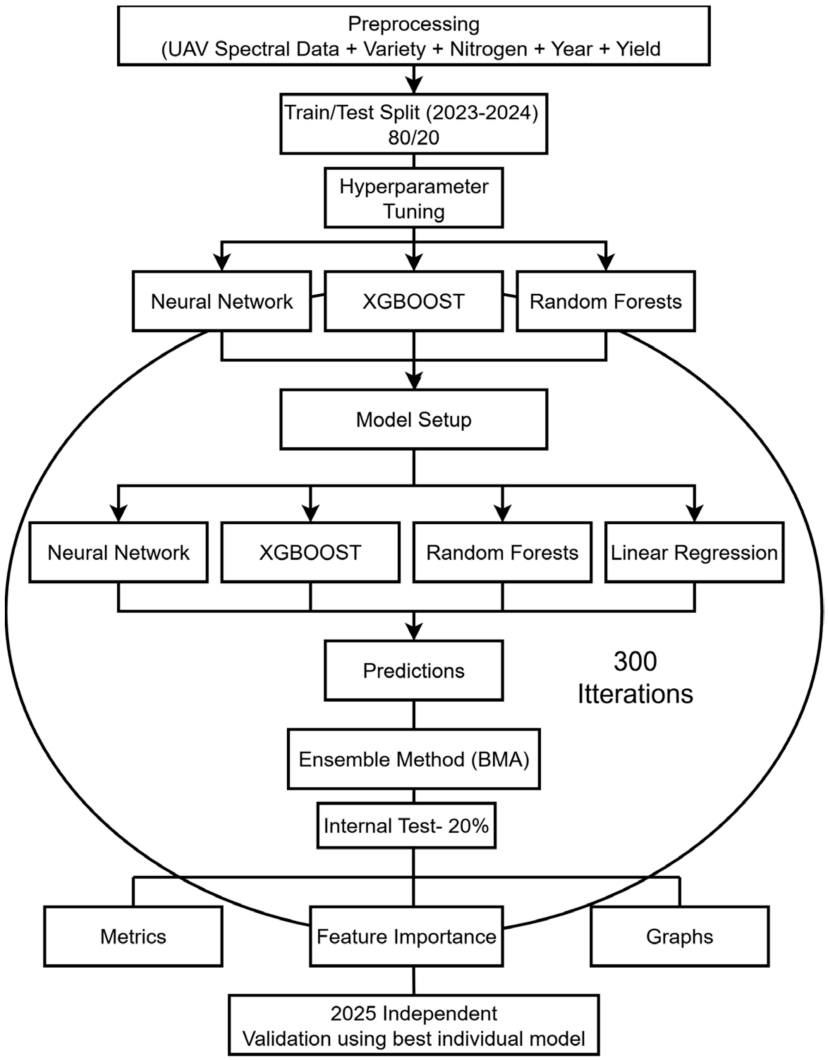

Figure 3. Modeling workflow for UAV-based rice yield prediction, showing data preprocessing, hyperparameter tuning, model training, and performance assessment.

The training dataset comprised 456 observations (individual plots) from the 2023–2024 growing seasons. For 2025 external validation (n = 120), year indicators were set to zero, preventing the model from relying on memorized year-specific patterns and forcing reliance on spectral features, variety identity, and nitrogen rates. This approach tests the true temporal generalization. All continuous features were standardized using z-score normalization (mean = 0, standard deviation = 1) prior to modeling to ensure an equal weighting across features. This step is crucial for distance-based algorithms and gradient descent optimization.

The 2023–2024 data (n = 456 plots) was partitioned into training (80%, n = 365) and internal testing (20%, n = 91) sets using stratified random sampling to maintain a proportional representation of varieties and nitrogen treatments. The training set was used for model development, hyperparameter tuning via 5-fold cross-validation, and stability assessments. The internal test set served as a holdout for within-distribution performance evaluation. The 2025 growing season data (n = 120 plots) was withheld entirely during model training

and served as an independent external validation set to assess temporal generalizability. All models were trained exclusively on 2023–2024 training data and subsequently evaluated on the internal test set (within-year performance), and the best individual model was evaluated in the 2025 external validation set (across-year robustness).

2.4. Machine Learning Models and Ensemble Methods

Four ML algorithms representing diverse learning paradigms were evaluated, along with a BMA ensemble implemented in two configurations (full and reduced) (Figure 3).

2.4.1. Random Forest (RF)

RF is an ensemble of decision trees that reduces overfitting through bootstrap aggregation and random feature selection [52]. Each tree in the forest is trained on a random subset of training data, and predictions are averaged across all trees. RF handles nonlinear relationships, feature interactions, and requires minimal preprocessing, making it robust for agricultural datasets [52]. In our implementation, we used scikit-learn (v1.3.0) RandomForestRegressor [53].

2.4.2. Extreme Gradient Boosting (XGBoost)

XGBoost constructs an ensemble of weak learners (shallow trees) sequentially, with each tree correcting residual errors from previous iterations [54]. The algorithm incorporates L1/L2 regularization, which prevents overfitting and uses a gradient-based optimization framework for efficient training. XGBoost has demonstrated a superior performance for tabular data in numerous agricultural applications [54].

2.4.3. Neural Network (NN)

A fully connected feedforward NN was implemented using TensorFlow (v2.13, Google LLC, Mountain View, CA, USA)/Keras, following a common application in agriculture yield prediction [55,56]. NNs model complex nonlinear relationships through successive transformations but are sensitive to dataset size and prone to overfitting without appropriate regularization [55]. The architecture was determined via Optuna Bayesian hyperparameter optimization (Section 2.5), balancing model capacity with overfitting risk given the dataset size (n = 456). The final network consisted of an input layer (47 features), four hidden layers (512, 1024, 704, and 576 neurons) with Leaky ReLU activation ($\alpha = 0.01$), and a single-neuron output layer predicting grain yield (t ha^{-1}).

Four complementary regularization strategies prevented overfitting. These included: (1) dropout (rates: 0.088–0.40, Table 5) forcing distributed learning, (2) early stopping based on validation loss, (3) batch normalization for training stability, and (4) 5-fold cross-validation ensuring generalization across data partitions. Further, performance stability across 300 iterations confirmed successful overfitting prevention.

2.4.4. Linear Regression (LR)

An ordinary least squares LR model was implemented as a baseline to establish a reference level of predictive performance under linear assumptions. Linear regression assumes additive and linear relationships between predictors and yield and therefore provides a lower-bound benchmark against which the value of more flexible, nonlinear ML models can be assessed [57]. Including LR allows evaluation of whether observed performance gains from tree-based and NN models arise from their ability to capture nonlinear interactions and complex feature dependencies, rather than from data leakage or overfitting. In addition, LR serves as a transparent benchmark commonly reported in yield prediction studies, facilitating a comparison with previous work and ensuring the interpretability of model performance differences.

Table 5. Hyperparameter search spaces and optimal values selected via randomized search with 5-fold cross-validation and Optuna. Grid ranges represent the bounds explored during optimization. Best parameters were determined by minimizing the validation set's RMSE. These optimized configurations were used for all subsequent model training and evaluation.

Models	Grid Range	Best Parameters
Random Forest	n_estimators: [100–900]	n_estimators: 100
	max_depth: [1–40]	max_depth: 21
	max_features: [1–60]	max_features: 31
XGBoost	n_estimators: [500–2900]	n_estimators: 1100
	learning_rate: [0.003–0.3]	learning_rate: 0.009
	max_depth: [5–30]	max_depth: 7
	subsample: [0.4–0.9]	subsample: 0.5
	colsample_bytree: [0.5–1]	colsample_bytree: 0.9
	gamma: [0–0.05]	gamma: 0.01
Neural Network	n_layers: [3–15]	n_layers: 4
	units per layer: [64–1024]	512, 1024, 704, 576
	dropout: [0.05–0.6]	dropout: 0.088
	learning_rate: [0.0001–0.01]	learning_rate: 0.00217
	activation	Leaky relu
	Optimizer	Adam
	Batch size [32,64,128]	Batch size 32
	Epoch [60–150]	Epoch 135

2.4.5. Bayesian Model Averaging (BMA)

BMA was employed to integrate predictions from multiple ML models while explicitly accounting for model uncertainty [39]. Rather than selecting a single best model, BMA combines predictions from a set of candidate models, thereby leveraging complementary strengths and improving robustness in predictive performance.

In classical Bayesian inference, BMA weights are derived from posterior model probabilities $P(M_k \mid D)$. However, in ML models, inverse-RMSE approximations are commonly used. This method emphasizes models with consistently lower errors and reduces the over-reliance on underperforming models.

The BMA ensemble prediction for observation i is defined as:

$$\hat{y}_i^{BMA} = \sum_{k=1}^{K} w_k \hat{y}_{i,k}$$

where $\hat{y}_{i,k}$ denotes the predicted yield from the kth model, and w_k represents the normalized model weight.

Two BMA configurations were evaluated to observe the influence of ensembles on predictive accuracy. BMA Full integrates predictions from all models used in this study, including both linear and nonlinear models, forming a comprehensive ensemble [39]

$$\hat{y}_i^{BMA}, \text{Full} = w_{RF}\hat{y}_i^{RF} + w_{XGB}\hat{y}_i^{XGB} + w_{NN}\hat{y}_i^{NN} + w_{LR}\hat{y}_i^{LR}$$

BMA Reduced excluded LR and combines only non-linear ML models. This reduced configuration was designed to test the hypothesis that inclusion of a linear model could dilute the ensemble performance [58].

$$\hat{y}_i^{BMA}, \text{Reduced} = w_{RF}\hat{y}_i^{RF} + w_{XGB}\hat{y}_i^{XGB} + w_{NN}\hat{y}_i^{NN}$$

2.5. Hyperparameter Tuning and Model Optimization

Hyperparameter optimization is crucial for optimizing model performance, as the default parameter values rarely yield optimal results across datasets and modeling tasks [49–51]. For RF and XGBoost, we employed scikit-learn's GridSearchCV, which performs an exhaustive search over predefined parameter grids [53]. GridSearchCV was selected for these tree-based models because their hyperparameter spaces are relatively low-dimensional and discrete (e.g., number of trees, maximum depth, and learning rate), making exhaustive grid search computationally feasible and ensuring comprehensive coverage of candidate configurations. The grid search was performed once with an inner 5-fold cross-validation on the training data of each split: the training portion was further split into 5 folds; the models were trained on 4 folds and validated on the 1 remaining fold, rotating through all folds; and the average validation RMSE was recorded for each hyperparameter combination. The combination yielding the lowest RMSE was chosen as the best. The tuned hyperparameters for each of the models are shown in Table 5.

Tuning the NN required the exploration of a larger and complex search space, and to address this, we used Optuna (v3.2), which performs adaptive, sequential optimization to efficiently evaluate high-dimensional hyperparameter combinations [59]. NN possesses substantially larger hyperparameter spaces involving continuous variables and architectural choices, making exhaustive grid search prohibitively expensive; Optuna's Bayesian optimization approach intelligently samples promising regions of the hyperparameter space, achieving efficient convergence with fewer evaluations [60]. Optuna proposed candidate architectures, each of which was trained and assessed using 5-fold cross-validation, and the trial with the lowest average validation RMSE was selected as the best configuration. This adaptive strategy enabled an exploration of complex architectural configurations while maintaining computational feasibility, ensuring that all models, despite different optimization methods, were tuned to their respective optimal performance levels for fair comparison.

The RF grid search identified an optimal model with a relatively shallow depth and low max_features, indicating that simpler trees and considering fewer features per split generalized better (likely due to noise in the high-dimensional feature space) [61]. Similarly, the XGBoost configuration favored a lower learning rate with a moderate number of trees and shallow depth. The best NN architecture discovered by Optuna had 4 layers (3 hidden layers) with a decreasing-then-increasing pattern in layer widths (Table 5). This architecture suggests that a moderate network with sufficient regularization provides the top performance. The selected learning rate balanced training speed and convergence stability.

2.6. Model Evaluation Framework

2.6.1. Performance Metrics

The coefficient of determination (R^2) measures the proportion of variance in observed yields, which is explained by the model, and it provides a standard indicator of model fitness, with higher R^2 values indicating stronger explanatory power [62].

The index of agreement (IOA) was used to assess and quantify the degree of agreement between observed and predicted yields. Unlike R^2, IOA is sensitive to both systematic and unsystematic errors, as it penalizes large deviations strongly [63]. IOA values range from 0 to 1, with values closer to 1 indicating stronger agreement between predictions and observations.

The A10 accuracy metric represents the percentage of predictions that fall within ±10% of the observed yield [64]. This metric provides an intuitive measure of model reliability, within commonly accepted agronomic error margins. Higher A10 values indicate a greater proportion of accurate predictions.

2.6.2. Error Metrics

These metrics were computed as follows. Let y_i denote the observed yield, $\hat{y}_i$ denote the predicted yield for observation i, $\bar{y}$ is the mean of observed yields, and n is the total number of observations.

Root mean square error (RMSE)

$$\text{RMSE} = \sqrt{\left(\frac{1}{n}\right)\sum_{i=1}^{n}(\hat{y}_i - y_i)^2}$$

Mean absolute error (MAE)

$$\text{MAE} = \left(\frac{1}{n}\right)\sum_{i=1}^{n}|\hat{y}_i - y_i|$$

Normalized root mean square error (NRMSE)

$$\text{NRMSE} = (\text{RMSE}/\bar{y}) \times 100$$

2.6.3. Iterative Validation for Stability Assessment

All models were evaluated across 300 random train–test splits (80/20) to ensure that the performance estimates were robust and not dependent on a single train–test partition [65]. The number of iterations (300) was determined through convergence analysis. Each iteration yielded a distribution of performance metrics, which enabled the calculation of mean, standard deviation, and coefficient of variation (CV) for these metrics, allowing us to quantify model stability. The convergence of the average cumulative RMSE was then monitored across iterations. The stabilization of the cumulative mean RMSE confirmed that 300 iterations were sufficient for a robust estimation.

2.7. Feature Importance

This study predicted rice grain yield using a set of 47 input features derived from a field experiment and UAV multispectral imagery. Feature-importance analysis was performed across 300 independent iterations to identify the most influential predictors and to assess their stability across modeling iterations. Model-specific importance measures were extracted from each algorithm; RF and XGB used impurity-based importance metrics derived from the Gini reduction, NN used permutation importance [52], and LR used the absolute value of its standardized coefficients.

To enable a comparison across models, importance scores were normalized within each iteration by dividing each score by the total importance across all features, yielding relative importance values [66]. An aggregate importance score for each feature was then obtained by averaging the normalized values across the four models. Features were ranked from 1 (most important) to 10 (least important within the top ten) based on these aggregate scores. Repeating this ranking procedure across all 300 iterations produced a rank distribution for each feature appearing in the top ten. Two stability metrics were derived: (1) the frequency with which each feature appeared among the top ten most influential predictors across 300 iterations, and (2) the distribution of its rank. The median rank represents the central tendency of each feature's importance across iterations, providing a robust indicator of its typical contribution to rice yield prediction.

3. Results

3.1. Yield Variation

Rice grain yield exhibited substantial variability across the three growing seasons (Figure 4, Table 6). In 2023, grain yield ranged from 2.98 to 14.87 t ha^{-1}, with a mean of 9.85 ± 2.65 t ha^{-1} (mean ± SD) and a coefficient of variation (CV) of 26.93%. In 2024, yields ranged from 4.93 to 12.72 t ha^{-1}, with a mean of 9.48 ± 1.55 t ha^{-1} and a CV of 16.38%. The 2025 growing season achieved the highest average yield (10.78 ± 1.74 t ha^{-1}), representing a 9.4% increase over 2023 and a 13.7% increase over 2024. Yield distributions were approximately normal in all seasons, with a greater dispersion observed in 2023 (CV = 26.93%) compared to 2024 and 2025 (CV ≈ 16%).

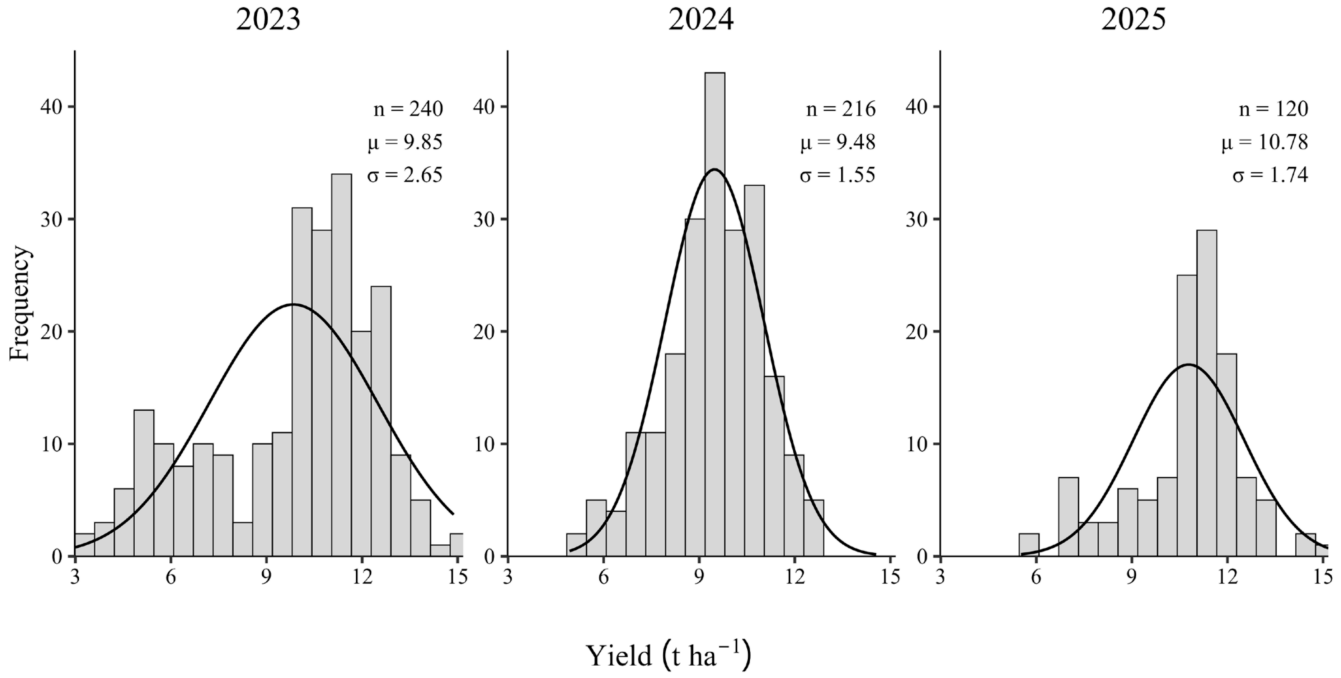

Figure 4. The frequency distribution of rice grain yield by growing season (2023–2025), with a fitted normal distribution curve.

Table 6. The descriptive statistics of yield per plot by growing season. The statistics include sample size (n), measure of central tendency (mean, median), dispersion metrics (standard deviation, coefficient of variation), range (minimum, maximum), and quartiles.

Year	n	Mean (t ha^{-1})	SD (t ha^{-1})	CV (%)	Min (t ha^{-1})	Q1 (t ha^{-1})	Median (t ha^{-1})	Q3 (t ha^{-1})	Max (t ha^{-1})
2023	240	9.85	2.65	26.93	2.98	7.74	10.54	11.67	14.87
2024	216	9.48	1.55	16.38	4.93	8.62	9.55	10.60	12.72
2025	120	10.78	1.74	16.10	5.52	10.40	11.10	11.80	15.20

Among the five varieties evaluated in 2025, CLHA03, the only variety tested in all three seasons, showed progressive yields of 10.50 t ha^{-1} (2023), 10.15 t ha^{-1} (2024), and 10.77 t ha^{-1} (2025). The four additional varieties evaluated in 2025, PVL04, Taurus, Fitzgerald, and 23DGL-003PV, increased from 9.15, 10.58, 9.20, and 9.22 t ha^{-1} in 2024 to 10.74, 11.25, 10.41, and 10.74 t ha^{-1}, respectively.

3.2. Model Training Performance and Stability

All four models were evaluated using 300 random train–test splits (80/20) based on the combined dataset from 2023 and 2024 (n = 456). The mean performance metrics and their variability across iterations are summarized in Table 7. XGBoost and RF exhibited the lowest CV (12.3% and 12.4%, respectively), followed by Linear Regression (13.8%). The NN showed the highest variability (CV = 15.2%).

Table 7. The model performance metrics across 300 random train–test iterations, presented as mean ± standard deviation and ranked by coefficient of variation (CV%) of RMSE.

Model	RMSE	R^2	MAE	CV (%)
XGBoost	1.20 ± 0.15	0.69 ± 0.08	0.87 ± 0.08	12.3
Random Forest	1.22 ± 0.15	0.69 ± 0.08	0.88 ± 0.08	12.4
Linear Regression	1.20 ± 0.17	0.69 ± 0.09	0.84 ± 0.08	13.8
Neural Network	1.40 ± 0.21	0.57 ± 0.16	1.08 ± 0.19	15.2

Cumulative RMSE convergence analysis indicated stabilization after approximately 200 iterations for all models (Figure 5). RMSE distributions across iterations showed narrower interquartile ranges and fewer outliers for XGBoost and RF compared to the NN (Figure 6).

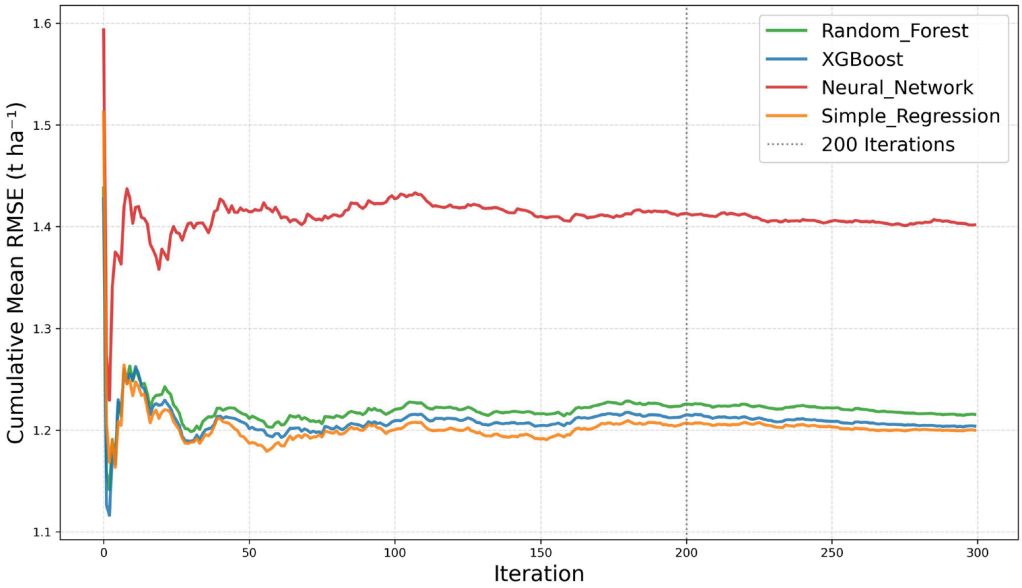

Figure 5. RMSE convergence over 300 random train–test iterations showing model stability. Lines represent cumulative mean RMSE for each model, calculated as the running average across all iterations.

While XGBoost and Linear Regression exhibited similar average RMSE across 300 iterations (both 1.20 t ha^{-1}), this similarity masks important differences in model robustness. Linear Regression showed higher performance variability compared to XGBoost, indicating less stable predictions across different data partitions. The higher coefficient of variation for Linear Regression suggests that its performance is more sensitive to the specific composition of training and test sets, whereas tree-based models demonstrate more consistent generalization regardless of data partitioning. This variability explains why performance on any single fixed test set (Section 3.3) may differ substantially from the 300-iteration average.

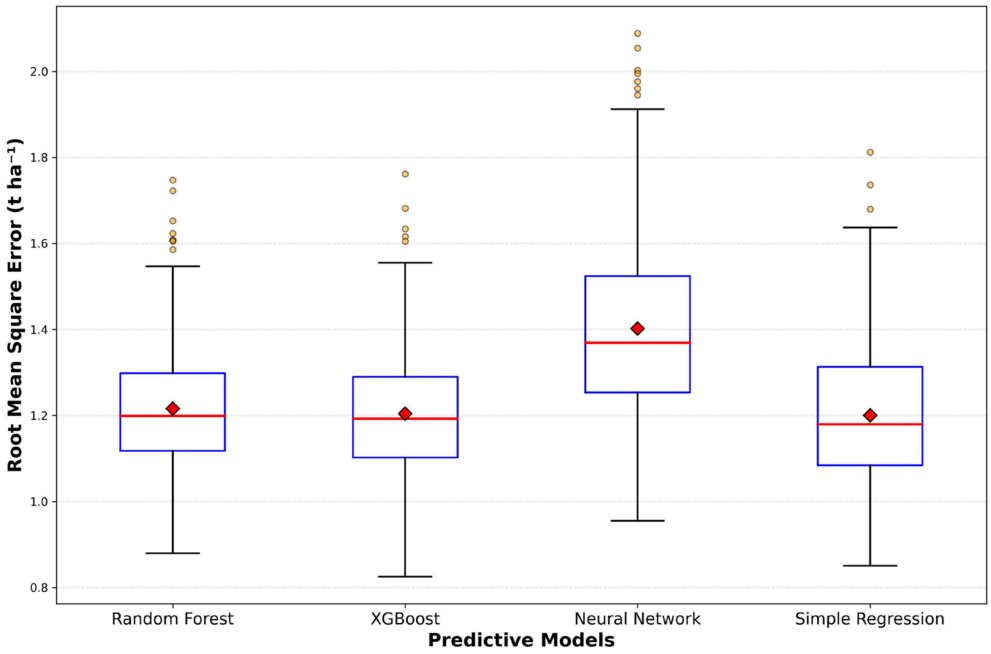

Figure 6. RMSE distribution across 300 random train–test iterations for four ML models. The box plots show the median (red line), mean (red diamond), and interquartile range (blue box). The whiskers extend to 1.5 × IQR, and the orange circles denote potential outliers beyond this range.

3.3. Model Performance on Independent Test Set

Final models were trained using optimized hyperparameters on the 2023–2024 training subset (80%, n = 365) and evaluated on an internal holdout test set (20%, n = 91). This single fixed partition represents one specific realization of the train–test split, and performance on this holdout set may differ from the 300-iteration averages reported in Section 3.2, particularly for models with higher performance variability. Performance metrics are presented in Table 8. The reduced BMA ensemble achieved the lowest RMSE (0.808 t ha^{-1}) and highest R^2 (0.886), followed by XGBoost (RMSE = 0.852 t ha^{-1}; R^2 = 0.873) and RF (RMSE = 0.911 t ha^{-1}; R^2 = 0.855). The NN achieved an RMSE of 0.978 t ha^{-1} and R^2 of 0.832, while Linear Regression showed the lowest performance (RMSE = 1.385 t ha^{-1}; R^2 = 0.664).

$$\hat{y}_i^{BMA}, \text{Full} = 0.257\hat{y}_i^{RF} + 0.260\hat{y}_i^{XGB} + 0.223\hat{y}_i^{NN} + 0.260\hat{y}_i^{SLR}$$

$$\hat{y}_i^{BMA}, \text{Reduced} = 0.348\hat{y}_i^{RF} + 0.351\hat{y}_i^{XGB} + 0.301\hat{y}_i^{NN}$$

Table 8. Final model performance on an independent test set (n = 91). Metrics include the coefficient of determination (R^2), the root mean square error (RMSE, t ha^{-1}), the mean absolute error (MAE, t ha^{-1}), the normalized RMSE (NRMSE, %), the index of agreement (IOA), and the percentage of predictions within ±10% of the observed values (A10, %). The model is ranked by RMSE.

Model	RMSE	R^2	MAE	NRMSE	IOA	A10
BMA Reduced	0.808	0.886	0.593	8.562	0.968	0.81
XGBoost	0.852	0.873	0.601	9.031	0.963	0.83
Random Forest	0.911	0.855	0.65	9.657	0.957	0.79
BMA Full	0.92	0.852	0.641	9.748	0.957	0.79
Neural Network	0.978	0.832	0.728	10.368	0.957	0.76
Linear Regression	1.385	0.664	0.875	14.683	0.904	0.66

Scatter plots of the observed yield versus predicted yields for all models are shown in Figure 7. Performance benefit indices and error metrics are summarized in Figures 8 and 9.

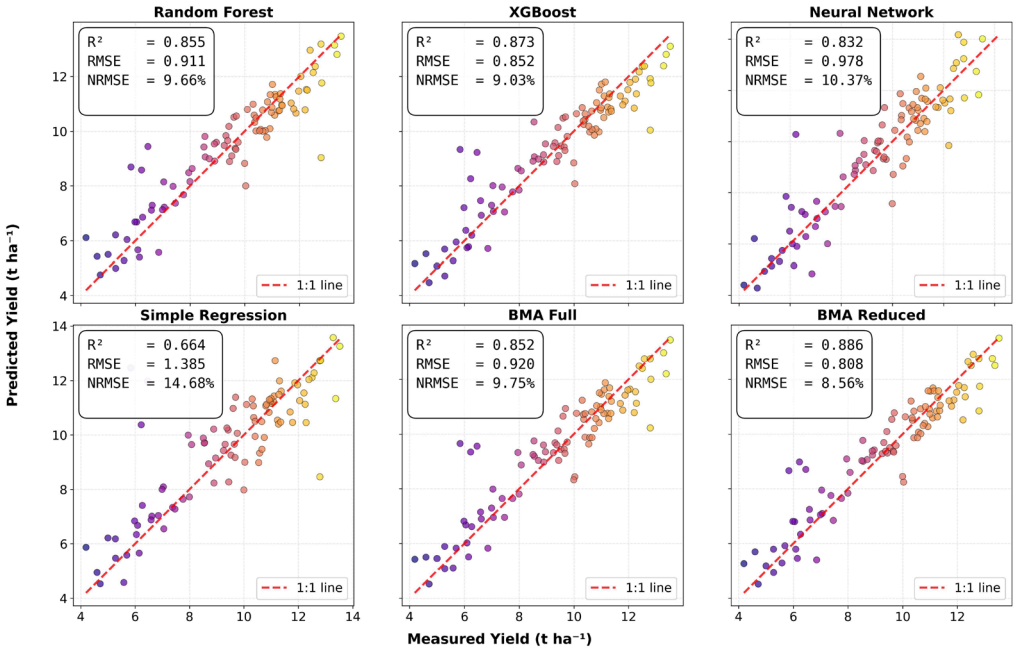

Figure 7. The scatter plots of the measured versus predicted rice yield for six models evaluated on the independent test set (n = 91). Each point represents an individual plot: the color gradient from purple (low yield) to yellow (high yield) has been provided for visual differentiation. Deviations from the 1:1 line indicate prediction errors; points above the line represent underpredictions, and the points below represent overpredictions.

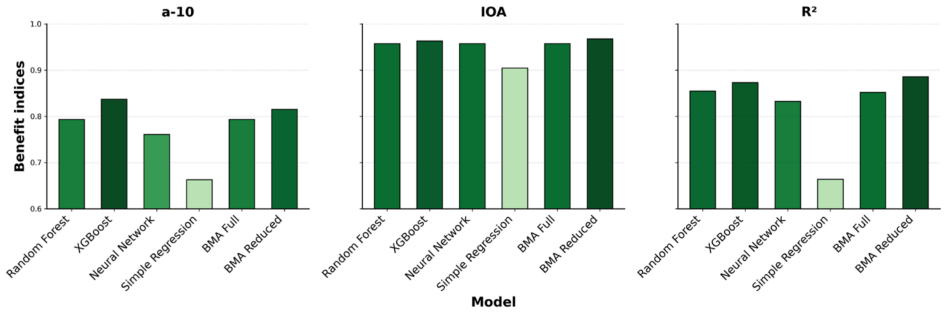

Figure 8. Benefit indices comparison R^2 (coefficient of determination), IOA (index of agreement), and A10 (predictions within ±10%) on the final test set. The higher values indicate a better performance.

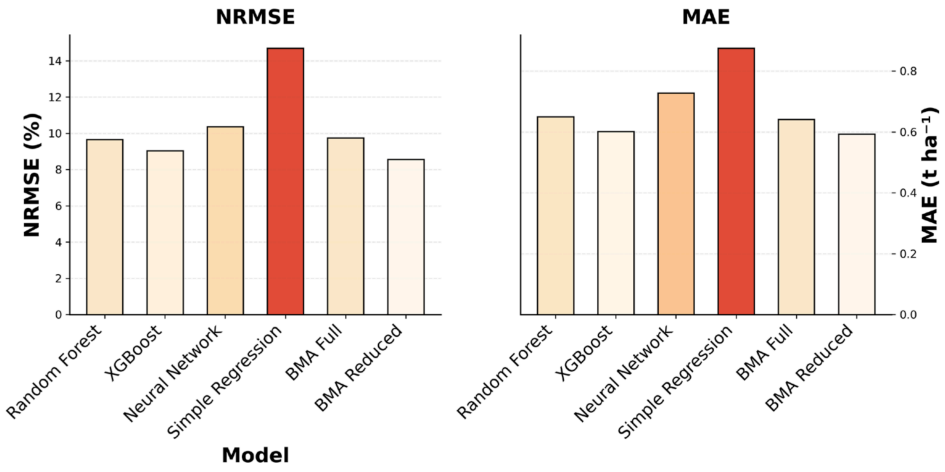

Figure 9. Error metrics comparison showing NRMSE and MAE on the final test set. The lower values indicate a better performance.

3.4. Feature Importance and Temporal Dynamics

Feature importance analysis across 300 iterations identified VIs derived at approximately 90 DAP as the most frequently ranked predictors of yield (Figures 10 and 11). Among all features, the EVI at 90 DAP appeared in the top 10 features rank 299 times out of 300 iterations with a median rank = 2, followed by the NDRE at 90 DAP (298/300 iterations; median rank = 2) and the GNDVI at 90 DAP (296/300 iterations; median rank = 3). These features showed high consistency across models and iterations. The NDVI at 90 DAP ranked fourth among VIs (frequency: 284/300 iterations; median rank = 4). In addition to VIs, individual spectral bands measured at 90 DAP are also ranked prominently. The near-infrared (NIR) band at 90 DAP appeared among the top 10 predictors in 192 out of 300 iterations (median rank = 6). Among the variety of indicators, DGM004 (V1), CLL19 (V11), and CLL18 (V4) appeared occasionally in the top 10 features, though with lower frequency and consistency.

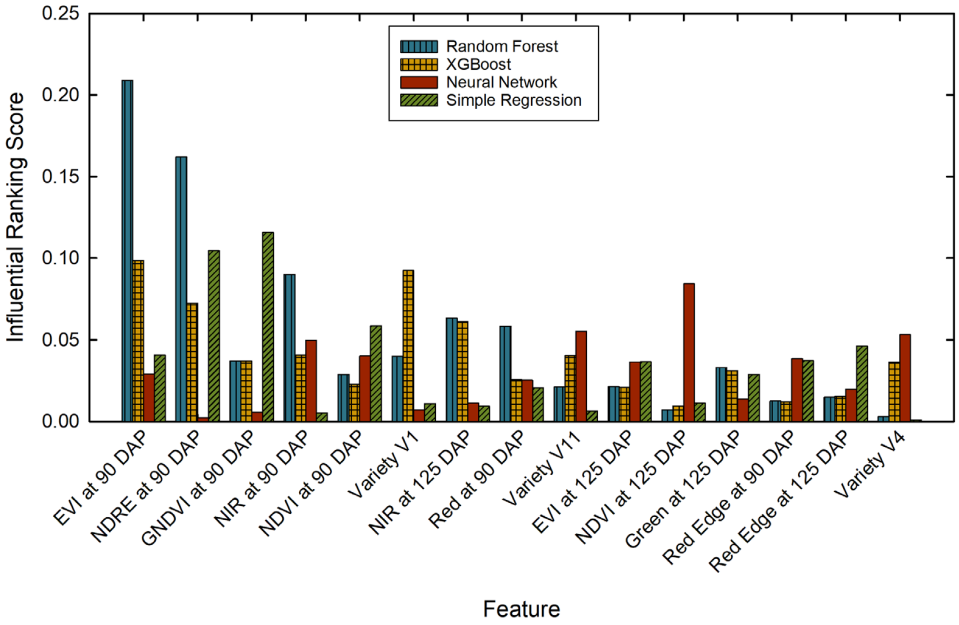

Figure 10. The topmost influential features for rice yield prediction. Normalized scores sum to 1.0 within each model.

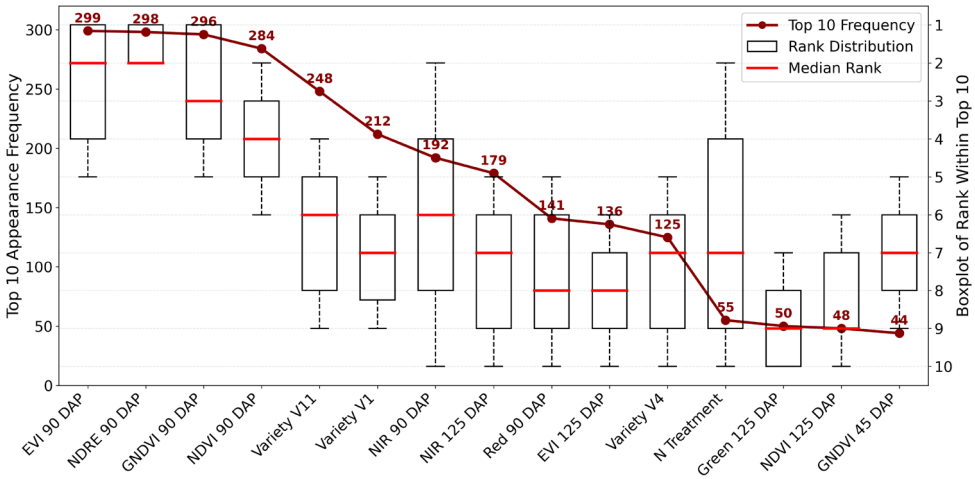

Figure 11. The stability and ranking of the top features for rice yield prediction across 300 model iterations. The left y-axis shows the frequency with which each feature appeared in the top 10 most important predictors out of 300 independent iterations. The right y-axis displays the distribution of rank positions for each feature when it appears in the top 10.

3.5. Independent Validation in 2025 Growing Season

The XGBoost model was evaluated on an entirely new 2025 growing season dataset (n = 120 plots) to assess model generalizability and perform independent validation. The performance metrics are presented in Table 9 and Figure 12.

Table 9. Final model performance on an independent test set (n = 120). The metrics include the coefficient of determination (R^2), the root mean square error (RMSE, t ha^{-1}), the mean absolute error (MAE, t ha^{-1}), the normalized RMSE (NRMSE, %), the index of agreement (IOA), and the percentage of predictions within ±10% of the observed values (A10, %). The model is ranked by the RMSE.

Model	RMSE (t ha^{-1})	R^2	MAE	NRMSE (%)	IOA	A10 (%)
XGBoost	0.907	0.624	0.725	8.473	0.87	78.57

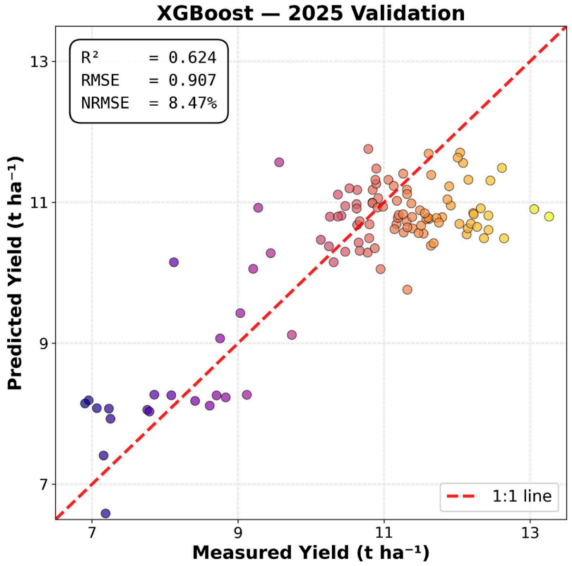

Figure 12. The measured vs. predicted rice yield in the 2025 independent validation set. Each point represents an individual plot: the color gradient transitions from purple (low yield) to yellow (high yield).

XGBoost achieved an R^2 of 0.624 (RMSE = 0.907 t ha^{-1}; NRMSE = 8.47%), and the index of agreement value was 0.869 for XGBoost, with approximately 79% of predictions within ±10% of measured yields (A10). The mean absolute error was 0.725 t ha^{-1} for XGBoost.

4. Discussion

4.1. Yield Variability and Experimental Robustness

The distribution of yield variability across the three growing seasons, with higher CV observed in 2023, reflects realistic field-scale variation typical of genotype × environment × management interactions. Similar interannual yield variability has been reported in multi-variety rice trials conducted under contrasting seasonal conditions (e.g., [40,42,67]), reinforcing the representativeness of the dataset used for model development.

The superior 2025 yield performance can be attributed to more favorable environmental conditions during critical reproductive and grain-filling stages. The 2025 growing season experienced minimal heat spikes during heading, with only 1 day exceeding the critical temperature threshold of 35 °C, compared to 11 days in 2023 and 2024, thereby minimizing the chance of spikelet sterility and maintaining high grain-setting rates in 2025. The consistent yield increase across all five varieties evaluated in 2025 confirms that environmental factors, rather than differential varietal composition, drove the 2025 yield advantage.

4.2. Comparative Performance of Machine Learning Models

Tree-based ensemble models (XGBoost and RF) consistently outperformed Linear Regression and the NN across both iterative train–test resampling and independent testing. This result is expected in UAV yield modeling because plot-level multispectral features are typically high-dimensional, correlated, and interact nonlinearly with the N rate and the genotype, and tree ensembles are well-suited to handle these properties without strong distributional assumptions (e.g., [54,68]).

XGBoost' s advantage is consistent with the design of gradient boosting as it builds an additive sequence of trees that iteratively correct residual errors, while using regularization and shrinkage to reduce overfitting [54]. RF, in contrast, reduces variance by aggregating many decorrelated trees built from bootstrap samples and randomized feature selection, which often yields robust performance under field heterogeneity and noisy measurements [52]. Collectively, these algorithmic properties explain why RF and XGBoost commonly perform well in UAV and remote-sensing yield prediction settings.

The comparatively weaker and less stable NN performance is also consistent with the established behavior of deep models in agronomic remote-sensing studies when sample sizes are modest [56,69]. NNs tend to require larger and more diverse training data to generalize reliably. A related example in rice shows that deep learning can achieve competitive error levels when trained on substantially larger multi-year datasets and richer inputs, which highlight the importance of scale and diversity for NN gains [41].

LR's consistently lower performance is explained by its core assumption that yield is modeled as a linear combination of predictors. However, in rice systems, yield responses to nitrogen and genotype are rarely purely linear; they involve thresholds, saturations, and interactions (G × E × M) that violate linearity. When relationships are nonlinear and interaction-driven, linear models generally underfit even if they remain interpretable [70].

The reduced BMA ensemble outperforms the full BMA, indicating that including a systematically weaker linear learner can dilute an ensemble if it receives non-trivial weight, especially when the underlying process is dominated by nonlinear effects [71]. This finding is consistent with ensemble learning theory that combining multiple models gives more accurate and reliable predictions [71].

The yield prediction accuracy achieved in this study (RMSE = 0.81 t ha^{-1}; NRMSE = 8.6%) is consistent with values reported in recent UAV-based rice yield prediction studies. For instance, our study estimates the yield at the individual plot scale using UAV multispectral data and ML across multiple rice varieties, nitrogen levels, and growing seasons, introducing a substantially greater genotype × environment × management variability. Under such conditions, reported RMSE values commonly range from approximately 0.8–1.5 t ha^{-1} [43,44,72]. Within this context, the performance of the reduced BMA and XGBoost models observed here is consistent with previously reported field-scale results and demonstrates the suitability of UAV-based ML approaches for high-resolution, in-season yield estimation to support precision rice management.

4.3. Generalizability and Model Robustness

XGBoost was selected for independent 2025 temporal validation as the best-performing individual model, providing a rigorous test of whether a single deployable algorithm learned robust spectral-yield relationships that are transferable across years. While Reduced BMA achieved marginally superior 2023–2024 performance (R^2 = 0.89 vs. 0.87), the improvement does not outweigh the operational advantages of validating a single model: computational efficiency, implementation simplicity, and conservative generalization assessment.

XGBoost maintained a strong predictive performance on the 2025 independent validation set despite substantial inter-annual variability, demonstrating robust temporal generalization. The modest decline in R^2 (~0.24 units) from internal validation to independent validation is consistent with expected performance degradation when models encounter novel environmental conditions not represented in training data [57]. Importantly, the NRMSE remained below 9%, which is considered suitable for operational decision support in precision agriculture applications.

The maintained predictive accuracy despite contrasting growing conditions provides strong evidence of genuine generalization rather than overfitting. Feature importance analysis confirmed that spectral indices (the EVI and NDRE at 90 DAP) dominated predictions. This demonstrates that the model relies primarily on spectral signatures to predict yields, with year capturing only residual environmental variation. The index of agreement (IOA = 0.87) and A10 metric (78.57% of predictions within ±10%) further confirm model reliability for operational yield forecasting applications.

4.4. Temporal Importance of Vegetation Indices

The strong predictive performance of features derived at approximately 90 DAP is physiologically plausible, as this period corresponds to late panicle initiation to early heading, when canopy structure and photosynthetic capacity are well established and reproductive processes governing yield are actively underway. Spectral measurements at this stage, therefore, capture integrated effects of the genotype, nitrogen management, and environment on yield formation [73]. While predictions at this stage offer limited scope for in-season intervention, they remain valuable for yield forecasting and post-season assessment of management strategies across fields.

The consistent performance of 90 DAP featured across 300 model iterations provides high confidence in the reproducibility of this temporal pattern. This stability suggests that operational UAV-based yield forecasting systems for rice could prioritize single-flight missions timed to coincide with the heading stage, reducing data acquisition costs and logistical complexity while maintaining high predictive accuracy.

4.5. Superiority of the EVI and Red-Edge Indices

Although the NDVI is commonly reported in crop yield prediction studies [14], in this analysis, the NDVI ranked fourth, being outperformed by the EVI and NDRE (Figures 10 and 11). This result is consistent with previous findings showing that the EVI and red-edge-based indices (e.g., the NDRE) are less prone to saturation under high-biomass conditions and less sensitive to soil background effects, making them more informative during dense canopy stages [50,74] (Figure 13). In contrast, the EVI at 90 DAP emerged as the single most stable and influential predictor. Its superiority can be attributed to its enhanced sensitivity under dense biomass and its incorporation of atmospheric correction and a soil-adjustment factor [74]. The NDRE at 90 DAP demonstrated nearly identical importance, highlighting the critical role of red-edge reflectance. Because the red-edge spectral region (705–750 nm) is strongly associated with leaf chlorophyll concentration and canopy nitrogen status, the NDRE captures variations in plant nutritional condition and photosynthetic capacity that are closely related to the final yield potential [75,76].

Raw NIR reflectance at 90 DAP (frequency 192/300, median rank: six) also ranked highly, reinforcing the association between higher NIR reflectance and healthier, denser canopies. Red-edge and NIR wavelengths have been identified as the most sensitive spectral regions for yield estimation [77], and their prominence in our analysis provides robust statistical confirmation.

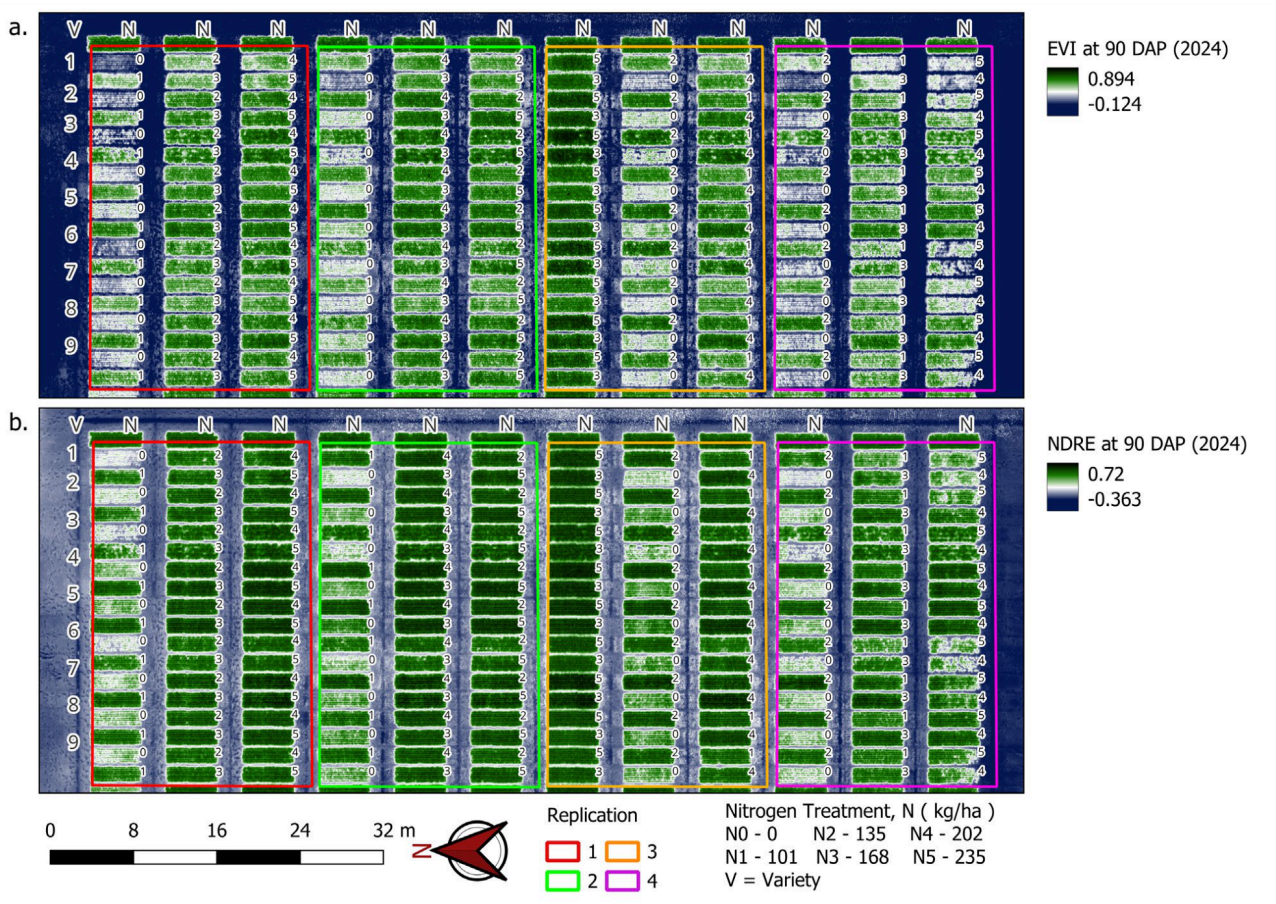

Figure 13. UAV-derived VIs maps from 2024 at 90 DAP, showing (**a**) the EVI and (**b**) the NDRE. Each replication block is delineated by a consolidated colored border (Rep 1–4), with N treatments (0–5) labeled within each plot.

4.6. Role of Varietal Information

The repeated appearance of varietal indicators among influential features (Figures 10 and 11) indicates that genetic differences in yield potential are not fully captured by spectral data alone. Three varieties, DGM004 (V1), CLL19 (V11), and CLL18 (V4), appeared occasionally in the top 10 predictors, but with substantially lower frequency and consistency than spectral indices (the EVI and NDRE at 90 DAP). This pattern is consistent with multi-genotype yield modeling studies, where explicit inclusion of varietal variation improved prediction [78], highlighting the importance of integrating genetic descriptors with remote sensing features when modeling yield across diverse rice germplasm.

4.7. Study Limitations and Future Directions

Despite the strong performance of tree-based and ensemble models, this study has several limitations. Although independent validation on 2025 data demonstrated model robustness across growing seasons, the dataset originates from a single location, which may limit model transferability across contrasting environments and soil types. In addition, the moderate sample size likely constrained the performance of the NN, as deep learning models generally require substantially larger and more diverse datasets [56,69]. Future research should prioritize multi-location field trials to assess model transferability, the expansion of training datasets to support deep learning approaches, and the integration of multi-sensor data (thermal, hyperspectral, LiDAR) to capture complementary physiological information. Explainable AI techniques should be implemented to improve model interpretability and diagnose model behavior across conditions. Emphasis should also be placed on developing

early-season yield prediction frameworks to strengthen the decision support for nitrogen management and climate-smart rice production.

5. Conclusions

This study presents a rigorous comparative evaluation of ML models for UAV-based rice yield prediction across three growing seasons with diverse environmental conditions. Tree-based models (RF, XGBoost) and BMA ensembles substantially outperformed Linear Regression, achieving up to 89% explained variance on internal validation and maintaining robust temporal generalization (NRMSE < 9%) on independent 2025 data despite inter-annual variation. Feature importance analysis across 300 iterations identified the EVI and NDRE at the reproductive stage (90 DAP) as the most influential and stable predictors of the rice grain yield. The framework demonstrates that spectral measurements at late panicle initiation to early heading provide reliable pre-harvest yield forecasting for operational precision rice production, supporting the field-level performance assessment and informing subsequent season management decisions. These findings advance the development of UAV-driven decision-support tools for site-specific crop management and establish a methodological benchmark for rigorous ML model evaluation in precision agriculture. Future research should evaluate model transferability across additional environments, seasons, and cropping systems.

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Conflicts of Interest: Author Rejina Adhikari is employed by Global Geospatial Institute. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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